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In Vivo Dissection of Two Intracellular Pathways Involved in the Spinal Oxytocin-Induced Antinociception in the Rat

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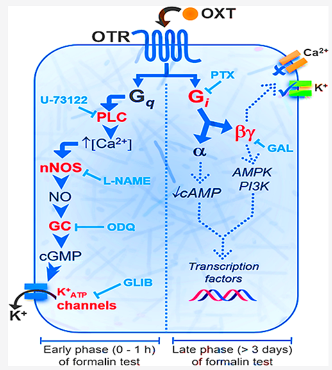
ABSTRACT: Behavioral and electrophysiological data show that at the spinal level, oxytocin inhibits pain transmission by activation of oxytocin receptors (OTRs). Canonically, OTRs are coupled to G*q* proteins, which induce a rise of intracellular Ca^{2+} by activating the phospholipase C (PLC). However, *in vitro* data showed that OTRs cause a plethora of intracellular events, some related to the activation of G*i* proteins. Using a behavioral approach, we analyzed the main *in vivo* intracellular pathway elicited by spinal oxytocin during a peripheral inflammatory/persistent nociceptive stimulus. Intrathecal oxytocin reduces early (number of flinches) and late (mechanical allodynia) formalin-induced nociception, an effect abolished by the OTR antagonist (L-368,899). Furthermore, the antinociception observed during the early phase (acute inflammatory) was also reverted by U-73122 (PLC inhibitor) but not by *pertussis* toxin ($G\alpha_{i/o}$ protein inhibitor) or gallein ($G_{\beta\gamma}$ subunit inhibitor). In contrast, the late oxytocin-induced behavioral analgesia was blocked by *pertussis* and gallein but not by U-73122. Since oxytocin's effects during the early phase were also antagonized by *Nω*-nitro-L-arginine methyl ester, ODQ, or glibenclamide (inhibitors of nitric oxide synthase [NOS], soluble guanylyl cyclase [GC], and K^+_{ATP} channels, respectively), the role of two differential pathways elicited by oxytocin is supported. Hence, we showed in *in vivo* experiments that oxytocin recruits two differential spinal intracellular pathways mediated by G*q* (PLC/NOS/GC/K^+_{ATP}) or G*i* proteins during a peripheral nociceptive stimulus.



KEYWORDS: *Analgesia, behavior, nociception, pain, oxytocin, spinal dorsal horn*

1. INTRODUCTION

The spinal dorsal horn is a primary site of action where peripheral nociceptive input can be modulated. Indeed, the neuronal transmission of painful peripheral stimuli can be blocked at the spinal cord level by an array of analgesic molecules activating specific receptors. In this context, oxytocin, a well-known neuropeptide associated with actions on lactation, uterine contraction, and social behavior, has also been related with spinal antinociception, and current data support the role of this nonapeptide as an endogenous pain killer at the spinal cord level.[1,2] Briefly, it has been shown that intrathecal oxytocin inhibits nociception in inflammatory[3] and neuropathic[4] pain models, and as recently reviewed by Poisbeau et al.,[2] electrophysiological and behavioral evidence supports the notion that activation of oxytocin receptors (OTRs) in the spinal dorsal horn induces analgesia, but how the activation of this G-protein coupled receptor produces this effect has been exiguously studied in *in vivo* systems.

Canonically, OTR is coupled to G*q* proteins, and *in vitro* assays from dorsal root ganglion cells showed that oxytocin elicits an intracellular increase of Ca^{2+} activating the nNOS/NO/K_{ATP} pathway.[5] However, *in vitro* cell systems also show that OTR couples to G*i* complexes;[6] thus, OTR activation could trigger differential intracellular pathways. Since after peripheral tissue injury, acute pain and long-lasting hypersensitivity can both be induced, a differential integration of central sensitization occurs.[7] Hence, we hypothesized that if oxytocin can recruit two differential intracellular pathways to inhibit early and late pain triggered by a peripheral injury, then pharmacologically blocking the downstream mediated by G*q* or G*i* pathways in an *in vivo* experiment helps us dissect if this peptide elicits a specific intracellular action. To test this hypothesis, and using the formalin test, we analyzed how blocking the OTR, phospholipase C (PLC, the main target of G*q* protein activation), the G*i* or $G_{\beta\gamma}$ proteins, or the nNOS/NO/K^+_{ATP} pathway disturbs the oxytocin-induced antinociception.

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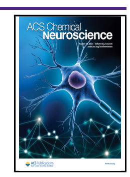

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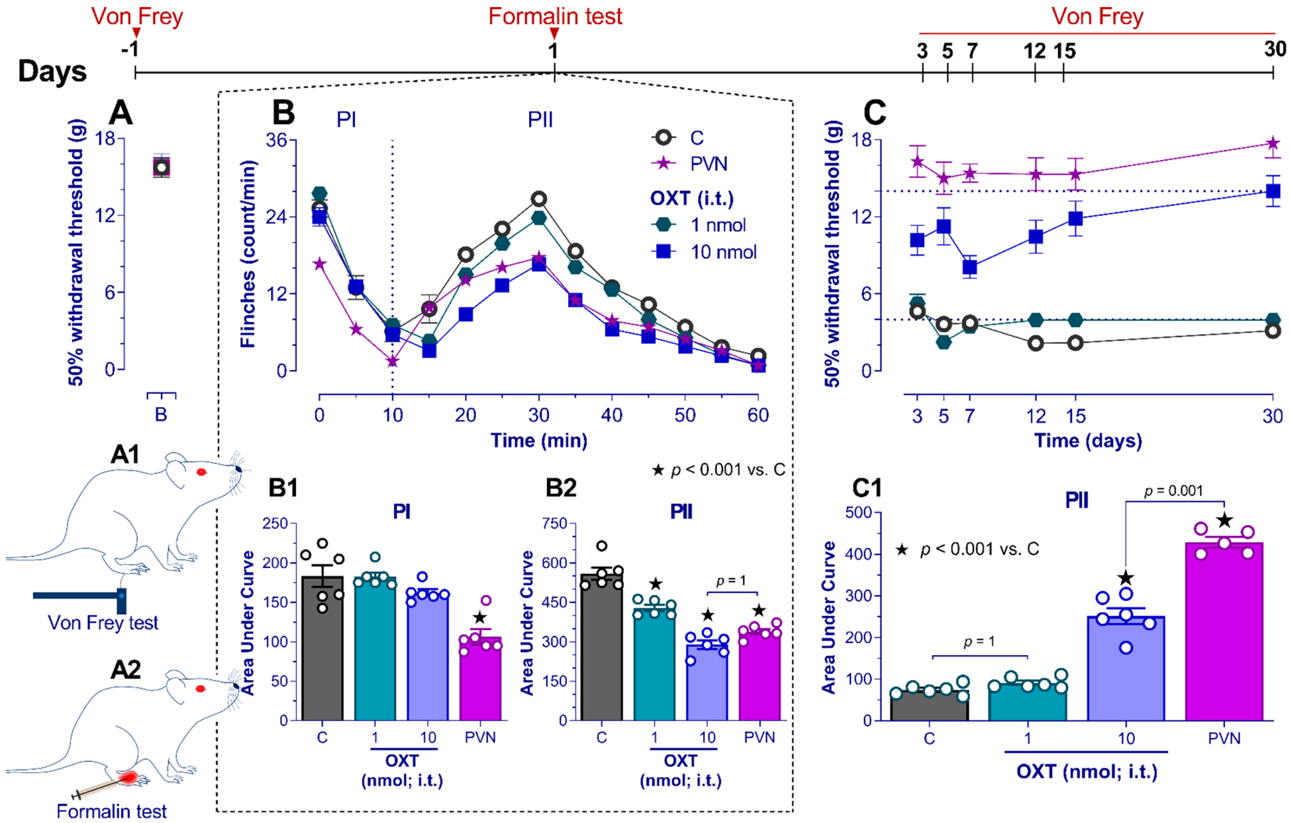

Figure 1. Spinal oxytocin (OXT) or electrical hypothalamic paraventricular (PVN) stimulation inhibits the early (0−1 h) and late (>3 days) nociception induced by formalin. Panel (A) shows the values of the paw withdrawal threshold (A1) measured 1 day before the formalin test (A2). In (B), formalin induced a characteristic biphasic time-course nociceptive behavior (PI and PII) expressed as flinches. In contrast to the control situation (○), intrathecal (i.t.) OXT diminishes the flinching behavior of PII, whereas PVN electrical stimulation reduces the flinching behavior in both phases. Panels (B1) and (B2) show the areas under the curve (AUCs) corresponding to flinching behavior during the time course of the formalin test. In (C), the time course of the paw withdrawal threshold induced by formalin over 30 days is depicted; note that 10 nmol of OXT or electrical PVN stimulation prevents the formalin-induced mechanical allodynia. Panel (C1) shows the AUC obtained from the late time course of the paw withdrawal threshold.

2. RESULTS AND DISCUSSION

2.1. Intrathecal Oxytocin Blocks Early (Flinches) and Late (Allodynia) Nociception Induced by Formalin. Subcutaneous formalin injection into the rat's hind paw evokes a well-known **early nociceptive biphasic response** (composed by phase I [PI] and phase II [PII]) behaviorally characterized as flinches in the injected paw (lasting approximately 1 h) and a **long-lasting hypersensitivity**, i.e., punctate mechanical allodynia evoked by a punctate stimulus with von Frey filaments (peaking at the third or fourth week) measured as a diminution of the paw withdrawal threshold.[8] We must acknowledge that the precise mechanisms that underlie the early and late formalin-induced nociception involve several pathways[8,9] that fall beyond the scope of the present work.

Using this model, we first investigated the effect of spinal oxytocin or hypothalamic paraventricular nucleus (PVN; the primary site of oxytocin synthesis) stimulation in the early (flinches) and long-lasting nociception (mechanical allodynia) induced by formalin. As shown in Figure 1, before formalin injection, the paw withdrawal threshold was nearly 16 g for all animals, indicating no hypersensitivity responses (Figure 1A). After formalin, a biphasic flinching behavior was immediately induced (the early flinching nociceptive response), and a drastic diminution of paw withdrawal threshold was observed over 4 weeks (the long-lasting nociception; allodynia). In animals treated with intrathecal oxytocin (1 or 10 nmol) or PVN stimulation (given 10 min prior s.c. formalin), the flinches (particularly during PII for oxytocin) and allodynia (along 4 weeks) were reversed (Figure 1B,C). These data agree with previous reports showing the antinociceptive effects of endogenous (by PVN stimuli) or exogenous oxytocin at the spinal cord level. Indeed, although oxytocin inhibits behavioral and electrophysiological nociceptive responses by OTR activation,[10−12] in the case of PVN stimulation, the role of supraspinal structures amplifying the antinociceptive effects during the formalin test cannot be disregarded.

For example, Motojima et al.[13] showed that subcutaneous injection of formalin elicits an increase of neuronal oxytocinergic activation at PVN. Similarly, Juif et al.[14] showed that intraplantar injection of λ-carrageenan increases oxytocin levels in the spinal cord and activates oxytocinergic neurons in the PVN. Moreover, PVN activation drives a direct descending spinal analgesic mechanism mediated by oxytocin[12,15] and could elicit an indirect descending noradrenergic activation by increasing the GABAergic activity in the rostral agranular insular cortex.[16] Under this evidence, to avoid unspecific nonspinal effects (and to simplify our experiments), we use exogenous oxytocin to pharmacologically dissect how oxytocin at the spinal cord level inhibits the early (flinches during PII) and long-lasting (mechanical allodynia) nociception.

2.2. Oxytocin-Induced Inhibition of Flinches and Allodynia by Recruitment of Two Differential Intra-

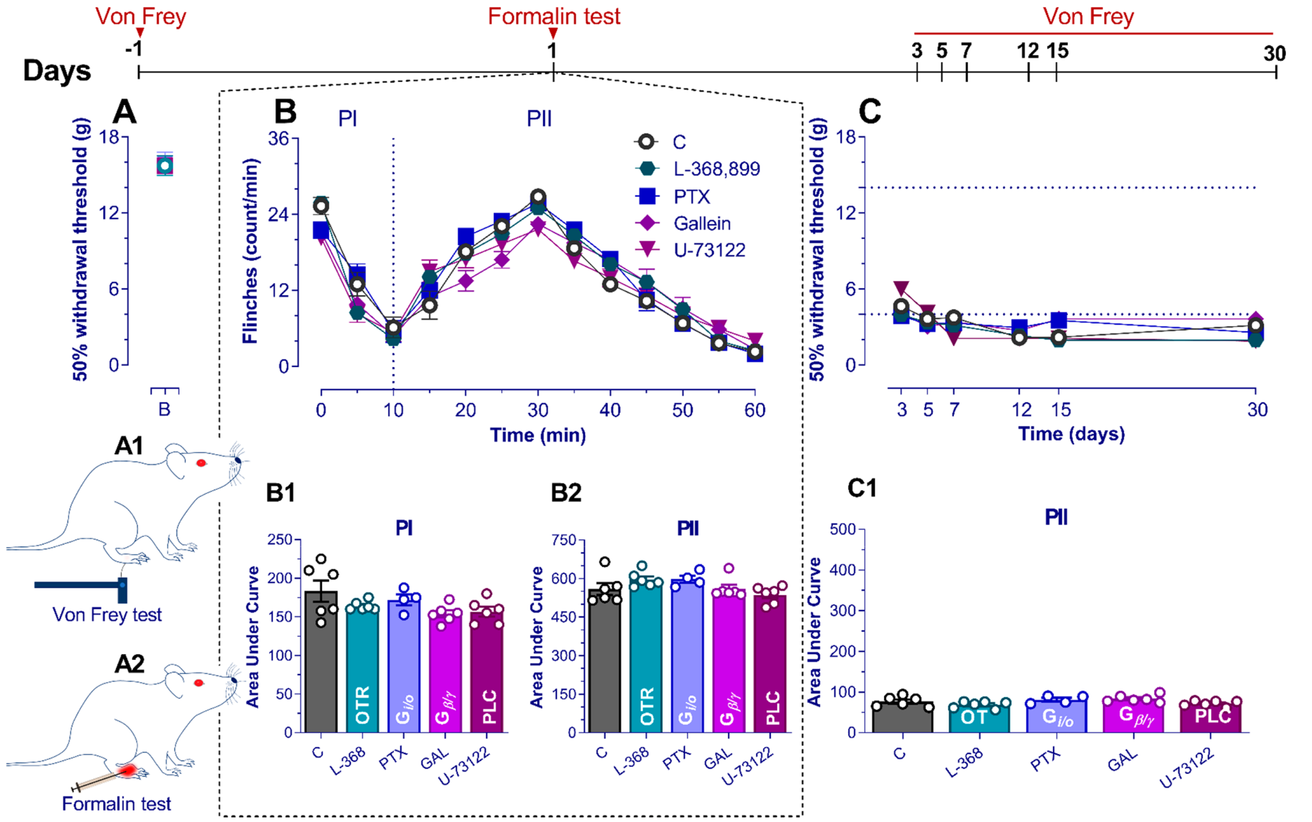

Figure 2. *Per se* effects of the OTR antagonist and inhibitors in early (0−1 h) and late (>3 days) nociception induced by 1% formalin. (A) Paw withdrawal threshold before the formalin test. (B) Time course during phase I (PI) and phase II (PII) of the mean number per minute of flinches observed after i.t. administration of the OTR antagonist (L-368,899), PLC inhibitor (U-73122), $G\alpha_i$ protein inhibitor (*Pertussis* toxin; PTX), or $G\beta/\gamma$ protein inhibitor (gallein; GAL) in rats submitted to the formalin test. The data show that the administration of L-368,899, U-73122, PTX, or GAL had no effect *per se* on the nociception induced by 1% formalin. (C) Time course of the paw withdrawal threshold in late nociception generated by 1% formalin. The data show that the compounds had no effect *per se* on the nociception induced by 1% formalin.

cellular Pathways Mediated by OTR Activation. Considering that OTR could drive intracellular signaling mediated by G*q* or G*i* proteins, several antagonist/blockers were used. Indeed, the activity of OTR, PLC, the $G_{\beta\gamma}$ subunits, or G*i* proteins was blocked using L-368,899 (a selective OTR antagonist), U-73122 (a PLC inhibitor), gallein (a $G_{\beta\gamma}$ subunit inhibitor), or *pertussis* toxin (an inhibitor of G*i* proteins). All treatments were given intrathecally 10 min before oxytocin (except for *pertussis* toxin, which was given 1 week before the formalin test). In this sense, we first tested the *per se* effect of the above blockers, and we found that these compounds do not affect the formalin-induced nociception (early and late) (see Figure 2).

Figure 3 shows that the oxytocin-induced antinociceptive effects at early (at PII; second phase of the formalin test) or late stages depend on two differential intracellular pathways. First, spinal pretreatment with L-368,899 inhibited the early and late antinociception induced by oxytocin. Second, U-73122 (but not *pertussis* toxin or gallein) inhibits the early (Figure 3B) but not the late (Figure 3C) effect. Third, the late antinociceptive effect of this neuropeptide was blocked by *pertussis* toxin or gallein (Figure 3C). Thus, the most parsimonious interpretation of these results is that the early and late antinociception induced by oxytocin is mediated by OTR but is mechanistically distinct. In the early stage, the role of G*q* proteins seems to be relevant, whereas in the late stage, the roles of G*i* and $G_{\beta\gamma}$ are necessary. Indeed, *in vitro* experiments using smooth muscle cells showed that OTR activation elicits muscle contraction by activating G*q* (via PLC activation)[17] and G*i* proteins (sensitive to *pertussis* toxin).[18] Furthermore, the BRET (bioluminescence resonance energy transfer) assay supports the contention that OTR promotes the engagement of both G*q* and G*i* proteins.[19]

Finally, it is interesting to point out that our results suggest that this neuropeptide could act as a pre-emptive analgesic. These results agree with a previous report showing that intrathecal injection of this neuropeptide prevents the development of postsurgical nociception.[20] Nevertheless, at this point, we do not know if the same mechanisms elicited to prevent the long-lasting nociception (i.e., allodynia) will be operative when the oxytocin is given as a curative treatment (when the allodynia is fully established). In any case, more experiments designed to tackle this question are necessary.

2.3. Role of the NOS/NO/cGMP/K^+_{ATP} Pathway in the Early Antinociception (Diminution of Flinches) Induced by Oxytocin. Considering that in cultured ganglion dorsal root cells this neuropeptide induces hyperpolarization by recruiting the Ca^{2+}/nNOS/NO/K^+_{ATP} pathway[5] and since the early oxytocin effect (diminution of flinches) depends on the activation of PLC (an enzyme that promotes intracellular Ca^{2+} mobilization) (see Figure 3B), we performed an experiment designed to indirectly test the *in vivo* contribution of the NOS/NO/cGMP/K^+_{ATP} pathway. First, we tested the *per se* effect of inhibitors and blockers of this pathway, L-NAME (a nitric oxide synthase [NOS] inhibitor), ODQ (guanylate cyclase [GC] inhibitor), and glibenclamide (K^+_{ATP} channel blocker),

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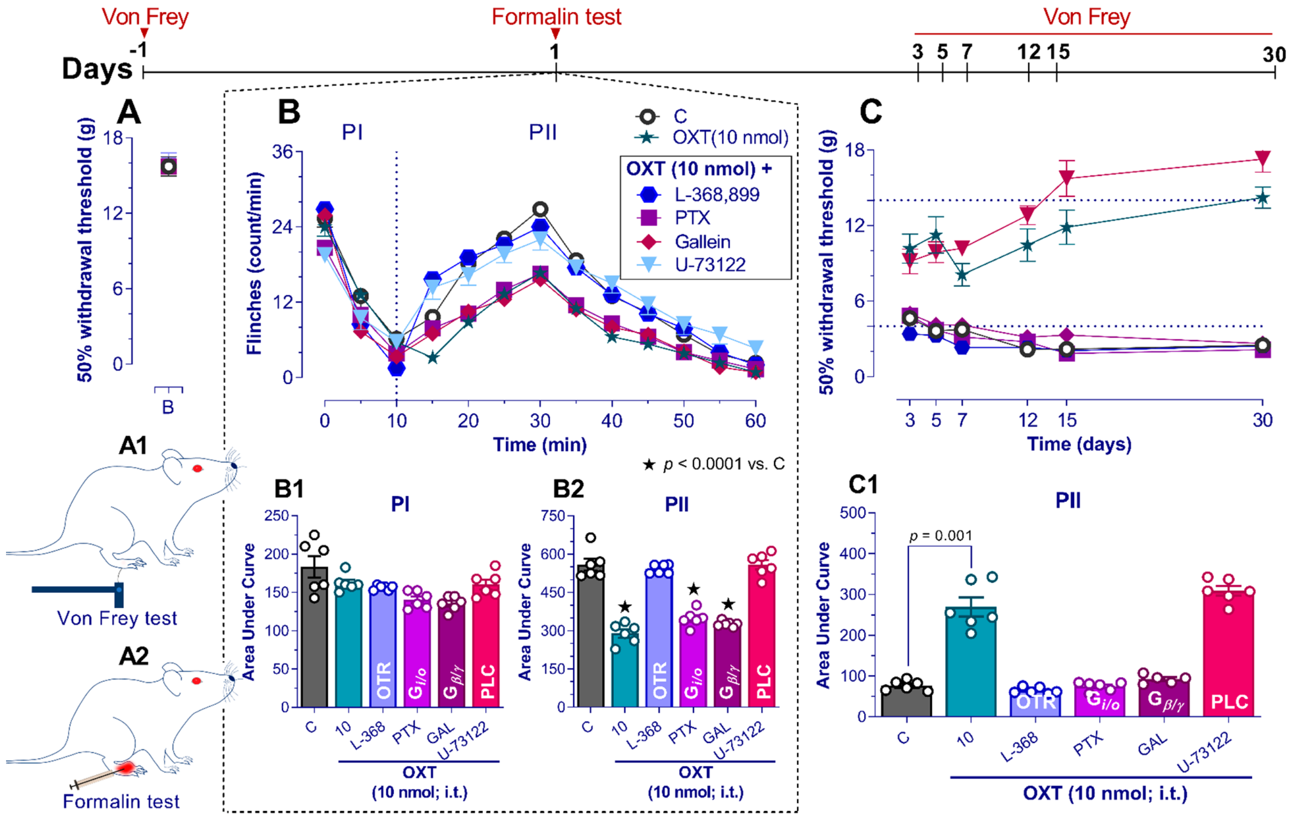

Figure 3. Oxytocin receptor (OTR) blockade inhibits early and late spinal oxytocin (OXT)-induced antinociception by two differential G-protein activations. (A) Paw withdrawal threshold (A1) was measured 1 day before the formalin test (A2). Panel (B) shows the effect of the pretreatment (before OXT) with (i) 10 nmol of L-368,899; (ii) 1 μg of *pertussis* toxin (PTX); (iii) 137 nmol of gallein (GAL); or (iv) 4 nmol of U-73122 on the time course during PI and PII of the flinches elicited by formalin; the control situation (○) and the effect of 10 nmol OXT (★) are shown for comparison. Panels (B1) and (B2) summarize the effect of the different treatments during the early nociception elicited by formalin; note that L-368,899 and U-73122, but not PTX or GAL, inhibited the effect of OXT, demonstrating the role of OTR and phospholipase C (PLC) mechanisms. In the late nociception induced by formalin, panel (C) shows L-368,899, PTX, and GAL block the OXT-induced late antinociception.

which did not modify the pronociceptive effect of formalin (Figure 4). Figure 5A shows that the early antinociception induced by oxytocin can be blocked when the animals were pretreated (given intrathecally 10 min prior oxytocin) with inhibitors of NOS, soluble GC, or K^+_{ATP} channels.

Indeed, it has been shown that PLC activation can generate antinociceptive effects, as described for the acetyl-L-carnitine, which through the activation of this enzyme, a reduction of the painful sensation is elicited in animals[21] and humans.[22] In addition, the NO/cGMP/K^+_{ATP} pathway is critical for the analgesic effects induced by current (e.g., opioids[23,24]) and potential analgesic drugs.[25]

2.4. Possible Locus of the Sensory-Inhibitory Effect Induced by OTR Activation. At this point, we must keep in mind that peripheral injury induced by an inflammatory stimulus can generate not only an immediate response but also spinal long-term potentiation (LTP), a phenomenon that partially explains the development of hypersensitivity after peripheral tissue injury.[26−28] Our experiments showed that spinal oxytocin blocks the initial nociceptive input (flinching behavior) and inhibits the long-lasting hypersensitivity; certainly, electrophysiological data showed that spinal oxytocin blocked spinal LTP induced by high-frequency electrical stimulation of the sciatic nerve.[11] Finally, our experimental design does not explain how or where the OTR is localized to induce antinociception. However, from previous studies (*in vitro* and *in vivo*), it could be inferred that OTRs are present in the primary afferent fibers,[4,11,12,15,29] exerting a presynaptic effect since the application of oxytocin at spinal cord level reduces the firing of Aδ- and C-fibers, without affecting the transmission of Aβ-fibers. Notwithstanding, other studies also raise the possibility that OTRs are also present in GABAergic interneurons.[14,30,31]

In conclusion, by analyzing the functional relevance of the main intracellular pathways elicited by spinal oxytocin during a peripheral tissue injury, we found that this peptide disrupts pain transmission by recruiting G*q* and G*i* proteins (Figure 5B). As discussed by Busnelli et al.,[6,19] OTR signaling is far more complex than a bimodal switch. Our *in vivo* experiments support the notion that OTR could activate several intracellular pathways mediated by different trimeric G-proteins in biased signaling. Thus, OTR activation triggers the G*q*/PLC/nNOS/cGMP/K^+_{ATP} pathway favoring the antinociception during the early phase, whereas for late antinociception, $G\alpha_{i/o}$ and $G\beta/\gamma$ protein activation are necessary.

3. METHODS

3.1. Animals and Husbandry. Experiments were performed on 106 adult male Wistar rats (180−220 g) provided by the bioterium of the Instituto de Neurobiología of the Universidad Nacional Autónoma a de México (INB-UNAM). The animals with free access to food and drinking water were housed in pairs in plastic cages with wood-based bedding, in a temperature (22 ± 2 °C) and humidity (50%) controlled room under a 12:12 h light/dark cycle (light beginning at 7:00 h). All procedures were approved by the INB-

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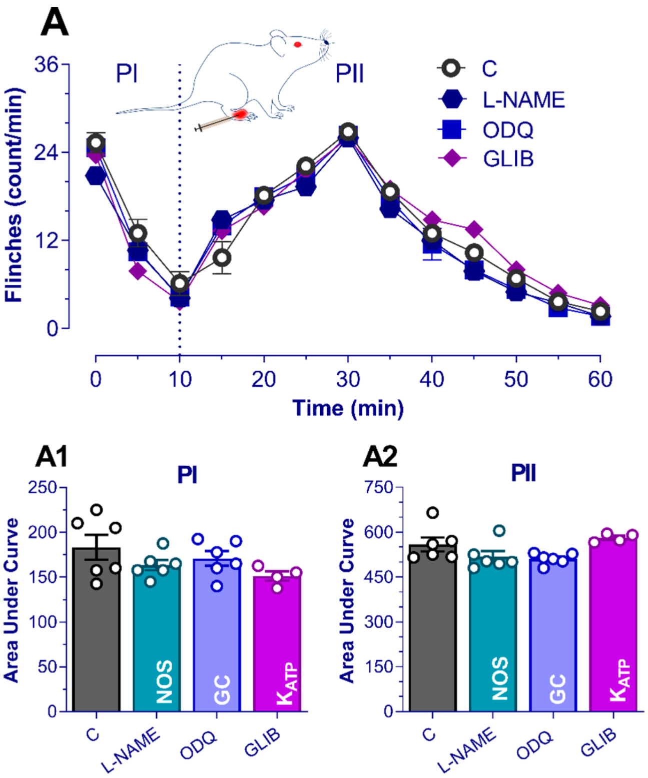

Figure 4. *Per se* effects of blockers of the NO/cGMP/KATP pathway in the early nociception (0−1 h) induced by 1% formalin. (A) Time course during phase I (PI) and phase II (PII) of the mean number per minute of flinches observed after i.t. administration of the NOS inhibitor (L-NAME), or GC inhibitor (ODQ), or ATP-sensitive potassium channel blocker (glibenclamide; GLIB) in rats submitted to the formalin test. The data show that L-NAME, ODQ, and GLIB administration had no effect *per se* on the nociception induced by 1% formalin in both phases of the formalin test (A1 and A2).

UNAM Ethics Committee following the NIH and ARRIVE guidelines. The number of rats used was the minimum to obtain reliable results, and efforts were made to minimize animal distress. All experiments were performed between 13:00−15:00 h. At the end of the experiments, the animals were halted in a CO_2 chamber.

3.2. Formalin Test. Nociception was assessed using the 1% formalin test.[32] Rats were placed in open Plexiglas observation chambers for 30 min for three consecutive days to allow them to become familiar with their surroundings. On the fourth day, and after 30 min in the Plexiglas chamber, they were removed for formalin administration. Formalin was injected subcutaneously (50 μL of diluted formalin, 1%) into the dorsal surface of the right hind paw using a 30G needle. Then, animals were returned to the chambers, and nocifensive behavior was observed immediately (**early nociception**). Early nocifensive behavior was quantified as the number of flinches of the injected paw during 1 min periods every 5 min, up to 60 min after formalin injection. Flinching was characterized as a rapid and brief withdrawal or flexing of the injected paw. Flinches induced by formalin administration generated a biphasic curve; the initial acute phase (0−10 min) was followed by a relatively short quiescent period, which was then followed by a prolonged tonic response (15−60 min). After acute nociceptive evaluation, animals were maintained for the next 30 days.

As previously reported by Fu et al.,[8] formalin induces not only an early inflammatory nociceptive response measured as flinches but also long-lasting hypersensitivity (**late nociception**). This hypersensitivity was assessed as a punctate mechanical tactile allodynia,[33] using the "up-and-down" method;[34] this method determines the mechanical force required to elicit a paw withdrawal response in 50% of animals. In short, rats were placed in individual transparent cages with a metal mesh floor and allowed to acclimatize for at least 30 min (on three different days) before the test. We used trials of 1.4, 2.0, 4.0, 6.0, 8.0, 10.0, and 15.0 g of Von Frey filaments (North Coast, USA) to estimate the 50% paw withdrawal threshold.

A series of filaments were presented in ascending order of strength to the plantar surface of the paw. Each hair was pressed (for 5 s) perpendicularly against the paw with sufficient force to cause slight bending. A positive response was noted if the paw was sharply withdrawn; at this point, we used the next filament to ensure that we had reached the threshold. If the response was negative, the filament size was increased to be sure of the threshold value. This paradigm continued until four more measurements had been made after the initial change of the behavioral response or until five consecutive negative (i.e., 15 g) or four consecutives positive (i.e., 0.25 g) responses had occurred. The scores were used to calculate the 50% withdrawal threshold using the statistical formula described by Chaplan et al.:[33] 50% g threshold = $10^{(Xf+\kappa\partial)}/10\,000$, where Xf = value of the final filament used; κ = tabular value of response (see Appendix 1 in Dixon[34]); and ∂ = mean difference between stimuli. This formula considers the number of positive and negative responses to have an indicator of the withdrawal threshold. Tactile allodynia was considered present with paw withdrawal thresholds ≤4 g.

3.3. Intrathecal (i.t.) Injection. As previously reported by Mestre,[35] a direct lumbar puncture in the L5−L6 intervertebral space was performed. All drugs given by i.t. injection were given in a 10 μL volume. This procedure requires an expert to perform the lumbar puncture; in this case, we measured the success index (~95%) to deliver i.t. lidocaine and observe a motor dysfunction of the hindquarters. The injections were performed holding the rat in one hand at the pelvic girdle level and inserting a 27G needle, connected to a 25 μL Hamilton syringe, in the i.t. space between the vertebrae L5−L6 on the dorsal side and perpendicular to the vertebral column. To do this procedure, the rats were briefly anesthetized in a hermetic box with a mixture of 6% sevoflurane (enriched with 1/4 O_2 and 3/4 N_2O); when motor reflexes were lost, a nose cone was used to maintain the anesthetic level (3.5% sevoflurane). The anesthesia outcome lasted less than 3 min.

3.4. Electrode Implantation in the PVN. Under anesthesia with ketamine/xylazine (70/mg/kg; i.p.), the animals were placed in a stereotaxic frame (David Kopf, USA), and a craniotomy was performed to place a concentric stainless-steel stimulation electrode (1 MΩ; Teflon-isolated microwire, Ø 279 μm, Cat. no. 7920; A-M Systems, Inc. USA) in the PVN (7.0 mm AP interaural, 0.2 mm lateral, and 2 mm height; Paxinos and Watson[36]). The electrodes and connectors were fixed to the skull using acrylic dental cement. Animals were allowed to recover from surgery for 6 days before formalin injection. The PVN stimulation consisted of a train of pulses of 1 ms at 60 Hz during a 6 s interval (S88 Stimulator, Grass Instruments Co., USA). The stimulation intensity (100−300 μA) was adjusted for each rat, depending on the behavioral responses and taking care to avoid side effects. The PVN stimulation was given 10 min prior to formalin administration.

3.5. Drugs. From Sigma Chemical Co. (St. Louis MO, U.S.A.), we used oxytocin acetate salt hydrate, 1-[6-[((17*β*)-3-methoxyestra-1,3,5[10]-trien-17-yl)amino]hexyl]-1*H*-pyrrole-2,5-dione hydrate (U-73122), *Nω*-nitro-L-arginine methyl ester hydrochloride (L-NAME), 1*H*-[1,2,4]oxadiazolo[4,3-*a*]quinoxalin-1-one (ODQ), 5-chloro-*N*-[4-(cyclohexylureido-sulfonyl)phenethyl]-2-methoxy-benzamide (glibenclamide), and *pertussis* toxin (PTX; *Bordetella pertussis*). From Tocris Bioscience (Northpoint, U.K.), we used (2*S*)-2-amino-*N*-[(1*S*,2*S*,4*R*)-7,7-dimethyl-1-[[[4-(2-methyl-phenyl)-1-piperazinyl]-sulfonyl]-methyl]-bicyclo [2.2.1]hept-2-yl]-4-(methylsulfon-yl)-butanamide hydrochloride (L-368,899) and 3′,4′,5′,6′-tetrahydroxyspiro-[isobenzofuran-1(3*H*),9′-(9*H*)xanthen]-3-one (gallein).

Oxytocin and PTX were dissolved in 0.9% NaCl solution. The L-NAME and glibenclamide were dissolved in 5% dimethyl sulfoxide (DMSO), and ODQ was dissolved in 10% DMSO, whereas gallein and U-73122 were dissolved in 1% DMSO. Moreover, the antagonist and/or blocker doses were high enough to block their respective targets in behavioral experiments.[37−43] The dose of oxytocin was

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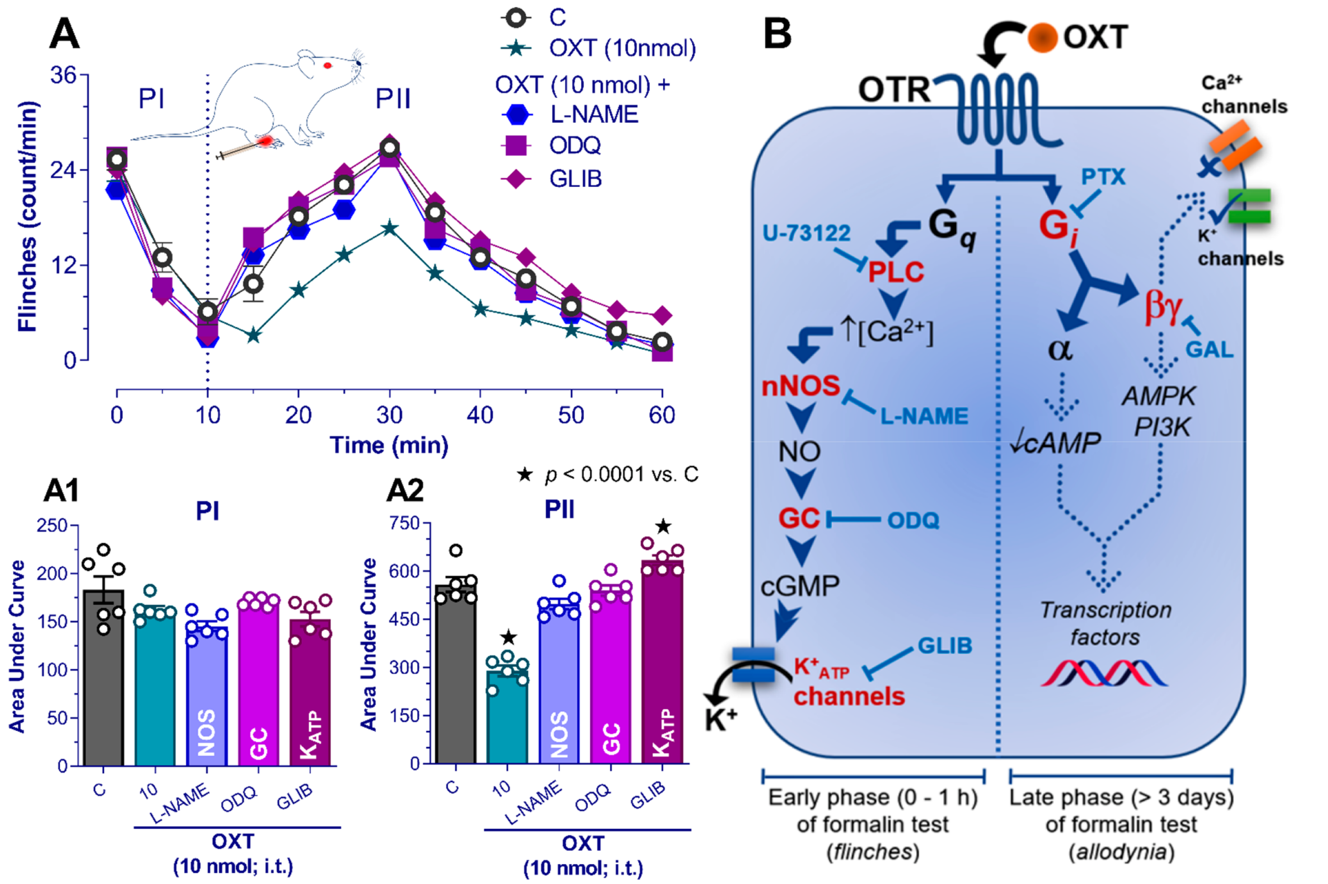

Figure 5. Role of the OTR/PLC/nNOS/K^+_{ATP} pathway in the early antinociception induced by oxytocin (OXT). In (A), the time course of different treatments during PI and PII of the flinches is depicted. The antinociception induced by 10 nmol of OXT (★; shown for comparison) was abolished by pretreatment with (i) 370 nmol of L-NAME (an inhibitor of nitric oxide synthase [NOS]); (ii) 50 nmol of ODQ (inhibitor of soluble guanylyl cyclase [GC]); or (iii) 100 nmol of glibenclamide (GLIB, an inhibitor of K^+_{ATP} channels). Panels (A1) and (A2) summarize the effect of the different treatments during the early nociception elicited by formalin; these data suggest that the early antinociceptive effect of OXT could be guided by the activation of the NOS/cGMP/K^+_{ATP} pathway. In (B), the proposed mechanisms involved in spinal OXT-induced antinociception in formalin-induced early and late nociception are depicted. This mechanism would start with the binding of OXT to its receptor (OTR). Once the OTR is activated, it would trigger two possible intracellular pathways. (1) For early antinociception, by coupling the OTR to a G*q*-type protein, the PLC would be activated, which induces the IP_3 release and consequently an increase of $[Ca^{2+}]_i$, recruiting the nNOS/NO/K^+_{ATP} pathway and promoting a hyperpolarization of the membrane. (2) For late antinociception, the role of a PTX-sensitive G*i* protein and the activation of the $\beta\gamma$ subunits seems to be relevant (the dotted lines indicate potential intracellular pathways involved in the antinociception induced by oxytocin).

chosen from previous reports showing that i.t. oxytocin inhibits nociception in a range of 0.1−10 nmol.[11,44−46] The vehicles and the antagonists did not affect the flinches or the paw withdrawal threshold (see Figures 2 and 4). Furthermore, oxytocin was given 10 min prior to formalin administration, whereas L-368,899, gallein, U-73122, L-NAME, ODQ, and glibenclamide were given 10 min prior to oxytocin administration. In the case of PTX, this toxin was given 1 week before the formalin test.

3.6. Analysis. All figures in the text are presented as the mean ± standard error mean (SEM) of the data obtained from six animals per group. The curves were constructed by plotting the number of flinches and as the 50% withdrawal threshold. The area under the curve (AUC) of the number of flinches and the withdrawal threshold was calculated using the trapezoidal rule. A reduction in the AUC of flinches elicited by formalin was considered antinociception, while in the 50% withdrawal threshold, an increase in the AUC was considered antinociception. The analysis of variance (ANOVA), followed by the Tukey test, was used to compare the differences between treatments. The differences were considered to reach statistical significance when the value of *P* was less than 0.01.

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Author Contributions

A.E.Z. and A.G.H. conducted the experiments and data analysis. All authors contributed to the writing of the manuscript and approved the final version that was submitted. M.C.L. and A.G.H. provided project administration and funding acquisition.

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Notes

The authors declare no competing financial interest.
Database deposition: Data obtained from the flinches evoked by formalin and the semiprocessed raw data related to the 50% paw withdrawal threshold over 4 weeks has been deposited in Mendeley Data.[47]
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ABBREVIATIONS

CGRP	Calcitonin gene-related peptide
GC	Guanylyl cyclase
L-NAME	$N\omega$-nitro-L-arginine methyl ester
LTP	Spinal long-term potentiation
NOS	Nitric oxide synthase
OTR	Oxytocin receptor
PLC	Phospholipase C
PVN	Hypothalamic paraventricular nucleus

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